FEDERATION UNIVERSITY AUSTRALIA · ITECH3208 IT CAPSTONE PROJECT 1

Sprint 4 Technical Report.

Pedagogy Evaluation Platform — a web-based self-assessment tool implementing Firmin's (2020) Techno-Pedagogical Practice framework. Sprint 4 delivered a user dashboard with three years of history, handling of current and previous attempts, historical comparison with charts, ten-minute evaluation sessions, and storage of evaluation data only on completion. The interface was redesigned and the production platform was recovered from an outage.

UNIT
ITECH3208 IT Capstone Project 1

SPRINT
Sprint 4

STATUS
Implemented, deployed and verified

DATE
14 September 2026
Revision 2 · 18 September 2026

LIVE SYSTEM
lmctpro2026.github.io/pedagogy-evaluation-platform

CLIENT
Dr Selena (Sally) Firmin

PROJECT TEAM

Mushfiqur Rahman	Lead Developer
Shafiullah Anik	Scrum Master
Albin Roy	Product Owner
Merajun Nabi	Tester

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01

Summary

Sprint 4 addressed six requirement areas raised by the client in sprint planning: a dashboard for signed-in users, storage and display of up to three years of evaluation data, correct handling of a current and a previous attempt (including the option to delete the previous one), year-to-year comparison presented with charts, a ten-minute limit on each evaluation session, and keeping an evaluation anonymous until it is completed.

All six are implemented, deployed to the live platform, and verified against the thirteen acceptance criteria derived from the client's wording (Section 13). The rules are enforced in the database rather than only in the browser, through a single write path for completed evaluations. The work also included a complete interface redesign and the recovery of the production database, which had become unreachable before the sprint began.

REQUIREMENT AREA	OUTCOME	STATUS
R1 · User dashboard	A dashboard page showing the active result, overall progress, stored history and every attempt.	DELIVERED
R2 · Three-year history	Attempts retained and displayed for three years; older data purged automatically.	DELIVERED
R3 · Current and previous attempts	Up to two attempts per year; the newest is the active result; the previous attempt can be deleted at submission or from the dashboard.	DELIVERED
R4 · Historical comparison	Per-category Improved / Decreased / Stayed similar labels, a comparison bar chart and a year-to-year line chart.	DELIVERED
R5 · Ten-minute session	Visible countdown with warnings; on expiry answers are cleared and nothing is stored; late submissions are rejected by the database.	DELIVERED
R6 · Anonymous until completion	No sign-in or database write while an evaluation is in progress; one transactional write on submission.	DELIVERED
Interface redesign	All seven pages rebuilt on a shared, accessible design system.	DELIVERED
Email delivery	Results email function hardened and made provider-independent; sender account configuration outstanding (Section 15).	OUTSTANDING

MEASURE	VALUE	MEASURE	VALUE
Acceptance criteria met	**13 of 13**	Production releases	**6**
Files changed	**32**	Lines added / removed	**6,836 / 3,393**
Automated database checks	**48, all passing**	New database functions	**3**

02

Sprint objectives

As specified by the client for Sprint 4.

REF	REQUIREMENT AS STATED
R1	Create a UI/dashboard for the user.
R2	Store and display evaluation data for up to 3 years.
R3	Handle two attempts/results. When there are two values/attempts, the system should handle the previous and current result properly. The user should have the option to delete/override the previous attempt. If the previous attempt is deleted/overridden, only the current attempt should be used as the active result.
R4	Allow users to compare their current results with previous years/past attempts, including category-wise progress. Show whether the user's teaching-practice scores have improved, decreased, or stayed similar. This can be displayed using graphs/charts on the dashboard.
R5	Each evaluation/session should have a 10-minute session time. The system needs to manage what happens when that session expires.
R6	The evaluation should remain anonymous while the session is in progress. Only after the session is completed should the system store the evaluation data.

INTERPRETATION AGREED DURING THE SPRINT

An early reading of R5 as an idle sign-out for account holders was replaced once the client's wording made clear that the ten minutes apply to each evaluation. An early reading of R3 as "one result per year, replaced automatically" was replaced by the client's "current and previous, with the option to delete the previous". Both earlier versions were superseded before release; the database migration was rewritten before it was run on the live system (Section 12, defect 1).

03

Sprint process

How the sprint was planned, tracked and closed.

The team works in Scrum with the client as the single source of requirements. Sprint planning took the client's six numbered requirement areas (Section 02) as the sprint goal and broke them into the backlog below. A story closed only when its acceptance criterion in Section 13 had evidence — a database test, a browser check, a network trace or a read-only probe against the live system. Two clarifications of the client's wording (R3 and R5, Section 02) arrived mid-sprint; both were absorbed before release rather than deferred.

SPRINT GOAL

Give returning participants a trustworthy view of their practice over time — current and previous attempts, three years of history, change by category — while keeping every evaluation private and unstored until it is complete.

Sprint backlog

#	STORY	ORIGIN	STATUS
S1	As a signed-in participant, I want a dashboard of my results so that my active result and history are in one place.	R1	DONE
S2	As a signed-in participant, I want my evaluations kept for three years so that I can look back across several years.	R2	DONE
S3	As a participant with two attempts in a year, I want the newest to be my active result and the earlier one kept as the previous attempt, so that a retake behaves predictably.	R3	DONE
S4	As a participant, I want to delete or override my previous attempt so that only my current attempt is used.	R3	DONE
S5	As a participant, I want to compare my current result with any past attempt, category by category, and see whether I improved, decreased or stayed similar.	R4	DONE
S6	As a participant, I want year-to-year change shown as charts on my dashboard.	R4	DONE
S7	As a participant, I want ten minutes per evaluation, with clear warnings and a defined outcome when time runs out.	R5	DONE
S8	As an anonymous participant, I want nothing stored until I submit, so that an abandoned evaluation leaves no trace.	R6	DONE
S9	As the client, I want the platform to look and read like a research instrument and to work on phones.	Added in sprint	DONE
S10	Restore the paused production database and apply the outstanding Sprint 3 and Sprint 4 migrations.	Unplanned (Section 04)	DONE
S11	As a participant, I want verification codes and results emails to arrive reliably.	Carried over	SPRINT 5

Definition of Done applied this sprint

- A rule is enforced in the database wherever one exists, not only in the browser.
- Automated database checks pass on a fresh schema and when applied over the previous migration.
- Behaviour is verified in the browser, including phone width and screen-reader announcements.
- The change is deployed to the live site and confirmed with read-only requests against the production database.
- The change is documented in this report with evidence against its acceptance criterion.

04

Production recovery

Context for the sprint: the live platform was not working when work began.

At the start of the sprint, sign-in and anonymous participation both failed on the live site. Investigation showed the fault was not in the application: the database host name returned `NXDOMAIN` from two independent DNS resolvers (Google and Cloudflare). The Supabase free-tier project had been paused after a period of inactivity, which removes its public address. The command-line access token recorded in the handover notes had also been revoked, so migrations could not be applied from the command line.

STEP	ACTION AND RESULT
Diagnosis	DNS lookups, authentication health and REST probes confirmed the project was unreachable rather than misconfigured.
Recovery	The project was restored from the Supabase dashboard and returned healthy (authentication settings unchanged: email sign-up and anonymous sign-in enabled).
Migrations	The outstanding Sprint 3 migration and the Sprint 4 migration were applied in order through the SQL editor (Figure 9).
Verification	Six read-only requests against the live database confirmed every server-side rule (Section 13).
Hardening	Sign-in and registration now report "Can't reach the sign-in service" instead of a raw "Failed to fetch" when the backend is unavailable.

> **LESSON FOR HANDOVER**
>
> Free Supabase projects pause after one week without activity. A scheduled keep-alive request and a regular data export are recorded as outstanding items (Section 15).

05

User dashboard and three-year history — R1, R2

Signed-in participants have a dashboard page (`my-results.html` , linked as "Dashboard" in the navigation). It is built from the participant's stored attempts, read under row-level security so each user can only ever load their own rows.

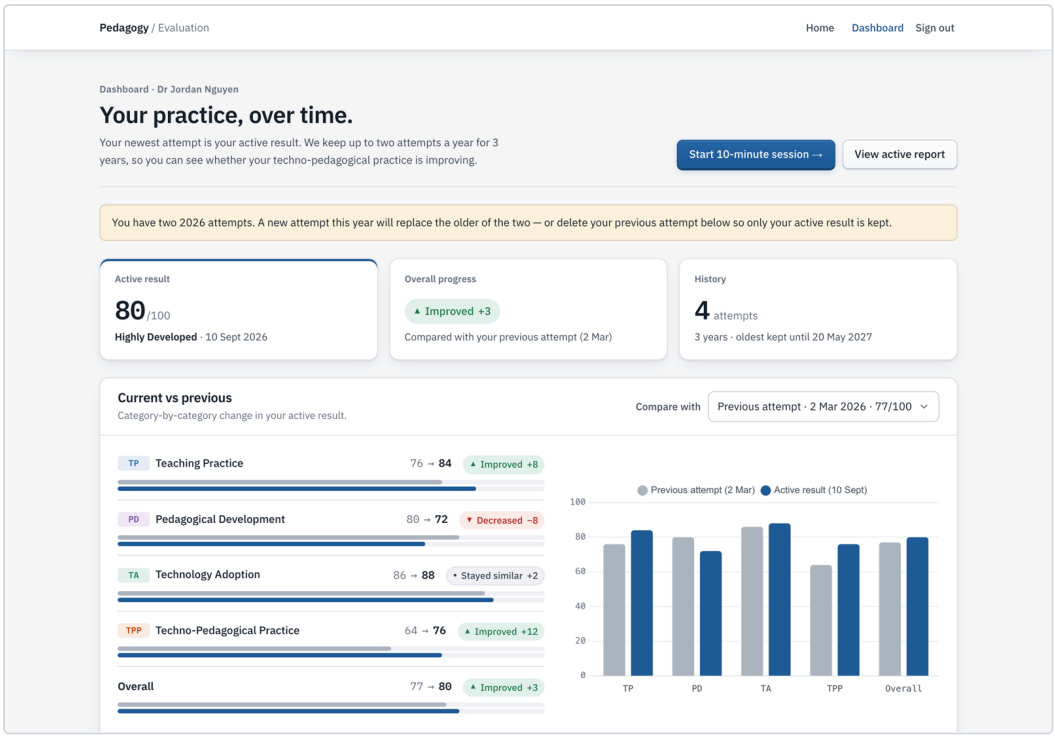


Figure 1. User dashboard: active result, overall progress against the previous attempt, and stored history. Test account with sample attempts.

Dashboard contents

AREA	CONTENT
Summary	Active result (overall score and descriptor), overall progress label against the selected comparison, number of attempts and the date the oldest will be removed.
Current vs previous	Selector for any earlier attempt; per-category rows with before and after scores, trend label and paired bars; comparison bar chart (Section 07).
Year-to-year progress	Line chart and table of each year's result for up to three years (Section 07).
All attempts	Every stored attempt with its status (Active result, Previous attempt, or year result), category scores, a link to its full report, and Delete for any attempt except the active result.
Other states	Signed-out, no attempts yet, and could-not-load messages, each with a clear next action.

Three-year retention

- The dashboard query only reads attempts completed within the last three years.
- The database function `purge_expired_assessments()` deletes any session older than three years. It runs on every submission and, where the `pg_cron` extension is available, daily at 03:15.
- Deleting a session cascades to its responses and demographics, so no orphaned answers remain.

06

Current and previous attempts — R3

RULE	BEHAVIOUR
Active result	The newest completed attempt is always the active result.
Previous attempt	The next most recent attempt in the same calendar year (Australia/Melbourne) is kept for comparison.
Limit	At most two attempts per year; submitting a third removes the oldest of that year.
Delete / override at submission	After a second attempt, the completion screen asks whether to keep both or delete the previous attempt.
Delete later	The dashboard offers Delete on any attempt except the active result.
After deletion	Only the current attempt remains active; admin averages count one attempt per participant per year.

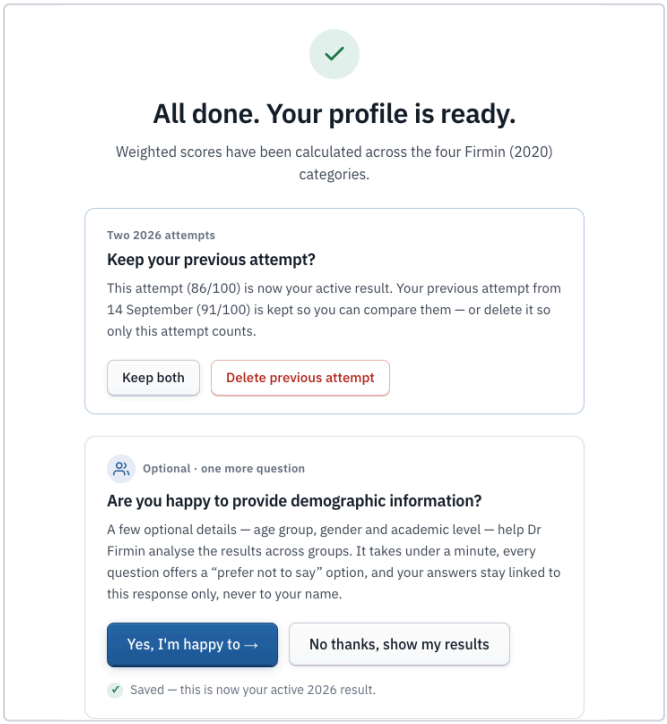


Figure 2. LIVE Completion screen after a second attempt on the live site with a real account: the new attempt (86) is active, the previous attempt (91) is kept, and the participant chooses to keep both or delete it.

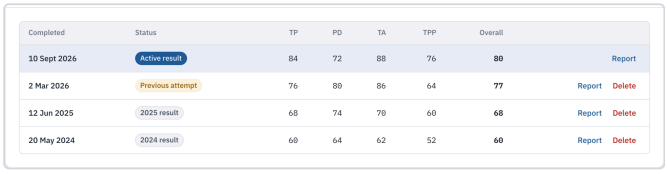


Figure 3a. Dashboard attempts before deletion: active result and previous attempt.

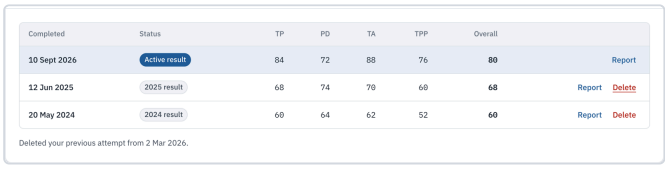


Figure 3b. After "Delete": the previous attempt is removed; only the current attempt is active.

Enforcement

Both rules are enforced by database functions, not only by the interface. `submit_assessment()` applies the two-per-year limit, and accepts a `p_replace_previous` flag that removes all earlier same-year attempts in the same transaction. `delete_previous_attempt()` deletes one of the caller's own attempts and refuses when that attempt is the caller's most recent — so the active result cannot be removed, whether by mistake or by a crafted request.

07

Historical comparison — R4

The comparison defaults to the previous attempt in the current year, or, if there is none, the most recent attempt from an earlier year. Any earlier attempt can be selected. Each category and the overall score receive one of three labels:

LABEL	RULE
Improved ▲	Score rose by more than 2 points.
Decreased ▼	Score fell by more than 2 points.
Stayed similar •	Change of 2 points or less, so small fluctuations are not presented as progress or decline.



Figure 4. Category-wise comparison showing all three labels — Teaching Practice improved (+8), Pedagogical Development decreased (−8), Technology Adoption stayed similar (+2) — with the comparison bar chart. Sample data.



Year	TP	PD	TA	TPP	Overall	Change from previous year
2026	84	72	88	76	**80**	▲ Improved +12 vs 2025
2025	68	74	70	60	**68**	▲ Improved +8 vs 2024
2024	60	64	62	52	**60**	First year on record

Figure 5. Year-to-year progress: each year's result (the newest attempt in that year) for up to three years, as a line chart per category and overall, with a table stating the change from the previous year. Sample data.

08

Ten-minute session — R5

Behaviour

STAGE	BEHAVIOUR
Start	The clock starts when the participant begins the evaluation. The start time is kept in the browser, so refreshing, closing the tab or opening a second tab does not reset it.
Running	The remaining time is shown in the progress bar throughout the questions and the review screen.
Warnings	At two minutes the timer turns amber and at one minute red; each threshold is also announced to screen readers.
Expiry	Unsubmitted answers are cleared from the device, nothing is stored or emailed, and an expiry screen offers a new session or a return home. A participant returning after the time has passed sees the same screen.
Server check	`submit_assessment()` rejects any submission more than ten minutes after the recorded start (plus one minute's grace for slow connections) with `SESSION_EXPIRED`, and rejects a missing or future start time.

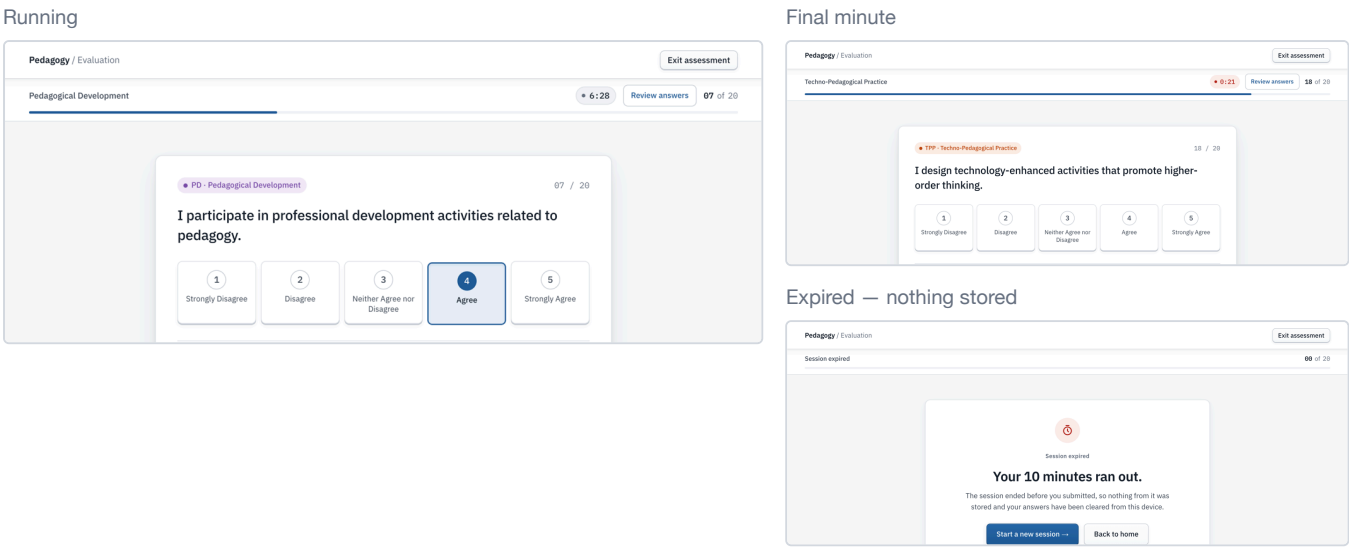


Figure 6. Session states: countdown during a question, the final-minute warning, and the expiry screen.

> **KNOWN LIMITATION**
>
> The start time is reported by the browser. The server check prevents ordinary circumvention (refreshing, reopening, submitting late), but a technically skilled user could alter the reported start time. This is appropriate for a reflective self-assessment; exam-grade enforcement would need a server-issued session token, which would conflict with R6 for anonymous participants.

09

Anonymous until completion — R6

Before Sprint 4, choosing "Continue anonymously" created an anonymous authentication account and a profile row, and beginning the questionnaire created a session row — so data existed before the evaluation was complete. Both behaviours were removed:

- Anonymous participants are not signed in to the backend at all; any existing sign-in on the device is ended first.
- No session row is created when an evaluation starts, for anonymous or signed-in participants.
- Answers remain only on the participant's device until submission.
- On submission, a single call to `submit_assessment()` stores the session and all twenty responses in one transaction, with no user attached for anonymous participants.

OBSERVED DURING A FULL ANONYMOUS RUN	NETWORK REQUESTS TO THE BACKEND
Answering all twenty statements	0
Review screen	0
Submission	1 database call (`rpc/submit_assessment`), then the results email request
Session expired before submission	0

10

Architecture and data model

The platform remains a static front end (HTML, CSS and vanilla JavaScript on GitHub Pages) over Supabase (PostgreSQL, authentication, edge functions). Sprint 4 moved the evaluation rules into the database so they hold regardless of the client.

Write path for a completed evaluation

```
submit_assessment(p_responses jsonb, p_started_at timestamptz,
                  p_replace_previous boolean default false,
                  p_participant_email text default null) returns jsonb

 1  reject a missing/future start time; reject if now − start > 10 min + 1 min grace
 2  require exactly the active questions (20), each answered 1–5; reject extra keys
 3  recalculate the four category scores from the answers (client scores are ignored)
 4  insert the session and all responses in one transaction
 5  signed-in only: remove abandoned in-progress rows; keep at most two attempts this
    year (or remove all earlier ones when p_replace_previous)
 6  purge sessions older than three years
 7  return { session_id, scores, year, previous_id, removed }
```

DATABASE OBJECT	PURPOSE
`submit_assessment()`	Only write path for evaluations. `SECURITY DEFINER`; executable by anonymous and signed-in roles.
`delete_previous_attempt()`	Deletes one of the caller's own attempts; refuses the most recent. Signed-in role only.
`purge_expired_assessments()`	Three-year retention. Not executable from the web roles.
`idx_sessions_user_completed`	Index on (user_id, completed_at) for history queries.
`handle_new_user()` **(revised)**	Account-creation trigger; now frees an email held by an orphaned profile row (Section 12, defect 6).

Client modules

FILE	RESPONSIBILITY
`js/session.js`	Participant identity, sign-out, and the ten-minute session clock (`PED.evalSession`). Clears a different participant's answers on shared devices.
`js/assessment.js`	Submission via `submit_assessment()` , attempt history, deletion; falls back to direct inserts if the migration is absent.
`js/questionnaire.js`	Timer display, warnings and expiry; attempt notes on the welcome screen; keep/delete choice on completion.
`js/my-results.js`	Dashboard rendering, comparison logic and charts (Chart.js).
`send-results-email`	Edge function: emails a results summary only for a session completed in the last 30 minutes, using stored scores.

Migration safety

`sprint4-attempts.sql` is idempotent and was tested both on a fresh schema and applied over its own earlier draft. A one-off clean-up keeps at most two attempts per user per year, removes abandoned in-progress rows

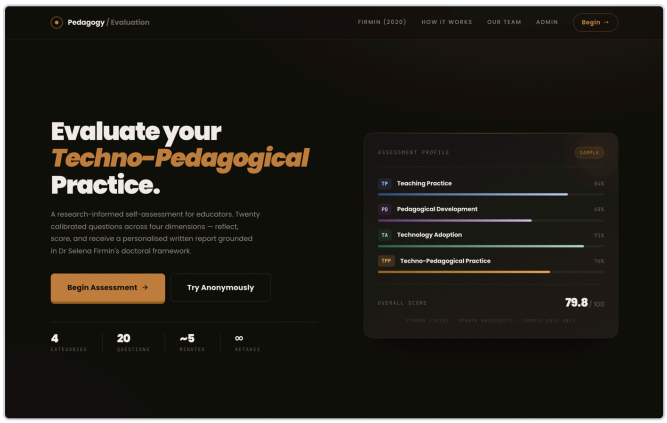
older than a day, and applies the three-year purge.

11

Interface redesign

The previous interface (dark theme, decorative animation, custom cursor, film grain, 3D card tilt) was visually striking but read as a template and was harder to use for a research instrument. All seven pages — home, questionnaire, demographics, results, report, user dashboard and admin — were rebuilt on a shared design system in `css/main.css`.

Before Sprint 4



Sprint 4

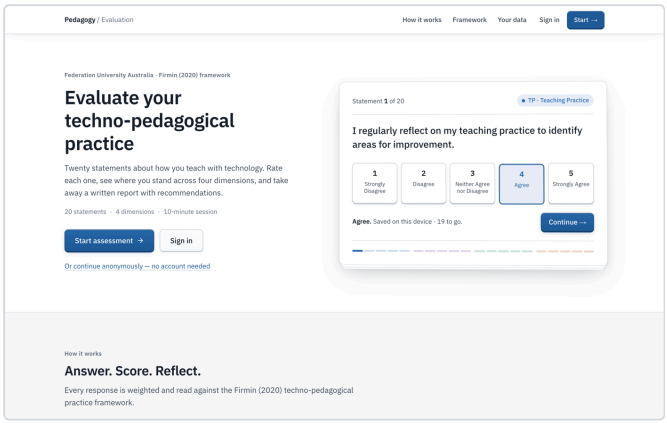


Figure 7. Home page before and after. The first statement can be answered directly on the home page and carries into the evaluation.

ASPECT	DECISION
Visual language	Light theme, one blue accent, restrained depth (layered shadows, tactile buttons and answer options).
Typography	IBM Plex Sans for text; IBM Plex Mono for scores, codes and the timer.
Colour meaning	The four framework category colours are used only to encode data (badges, charts).
Accessibility	WCAG AA text contrast, visible keyboard focus, skip link, reduced-motion support, layouts verified at 390 px width.
Removed code	Custom cursor script, GSAP animation library, 3D tilt, decorative gradients and grain.
Admin	Shares the site's tokens and components; averages count only each participant's active attempt per year.

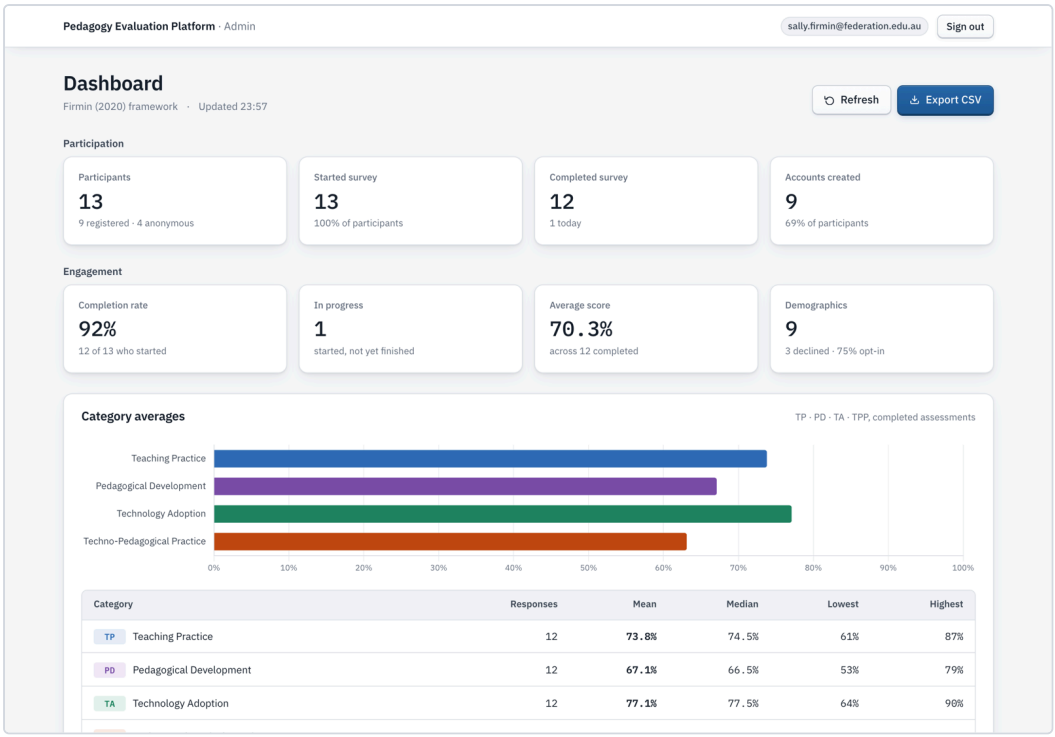


Figure 8. Redesigned administrator dashboard. Sample data.

12

Quality assurance

Defects found during development and review, all fixed and re-verified before release.

1 Draft migration would have deleted same-year attempts

`Severity: high · Area: database`

The first version of the attempt migration replaced a user's same-year result automatically, and its clean-up step deleted all but one attempt per year — contrary to the refined R3. It was rewritten before being run on the live database, and the new version was tested applied over the old one.

2 Anonymous participation wrote data before completion

`Severity: high · Area: authentication, persistence`

Starting anonymously created an authentication account and profile row, and beginning the questionnaire created a session row, contrary to R6. Removed; verified with a network trace (Section 09).

3 Results email function could send arbitrary email

`Severity: high · Area: edge function`

The function accepted scores and a recipient from any caller, so it could be used to send emails with fabricated content to any address. It now sends only for a session completed in the last 30 minutes, uses the scores stored in the database, and escapes participant-supplied text.

4 Admin averages could count a participant twice

`Severity: medium · Area: admin dashboard`

With a previous attempt kept, both attempts entered the category averages. The admin dashboard now counts only the newest attempt per participant per year.

5 Home-page statement card overflowed on phones

`Severity: medium · Area: responsive layout`

At 375 px width the fifth answer option was cut off because grid items would not shrink below their content width. Fixed and verified on a 390 px viewport.

6 "Database error creating new user"

`Severity: medium · Area: authentication trigger`

Creating an account failed when an old profile row (whose account had been deleted) still held the same email, violating the unique email constraint. The account-creation trigger now frees the email from orphaned rows. Reproduced and fixed in a test database (`fix-new-user-trigger.sql`).

7 "Continue" button visible before answering; unclear outage errors; wrong category label in emails

`Severity: low · Area: interface, email`

A stylesheet rule overrode the `hidden` attribute; sign-in showed "Failed to fetch" during outages; emails labelled PD as "Professional Development". All corrected.

13

Testing and acceptance

Database — 48 automated checks

The migration was executed with PGlite (PostgreSQL compiled to WebAssembly) against the full schema chain — `supabase-setup.sql` , `admin-policies.sql` , `sprint3-demographics.sql` , `sprint4-attempts.sql` — with mocked authentication functions for anonymous, signed-in and anonymous-token callers. The suite ran on a fresh database and over the earlier draft of the migration (24 checks each, every migration applied twice). All passed. Coverage: storage on completion only; expired, future and missing start times; grace period; incomplete, out-of-range, non-integer, string and extra answers; two attempts per year with the oldest removed; replace-previous; deleting a previous attempt; refusal to delete the active result; refusal across users; cascade of responses; three-year purge; permission denied on the purge function for web roles.

Browser — end to end

- Dashboard with full history, a single attempt, and no attempts; switching the comparison; all three trend labels; deleting a previous attempt.
- Timer at start, at two minutes, at one minute, expiry during a question and expiry on return; screen-reader announcements.
- Anonymous run with request monitoring (Section 09); completion-screen keep/delete choice.
- Sign-in behaviour with the backend unreachable; mobile layout at 390 px; 18 page states captured with no JavaScript errors.

Live system

REQUEST TO THE LIVE DATABASE (WRITES NOTHING)	RESULT
Submit an incomplete evaluation	Rejected — "All 20 questions must be answered"
Submit a session started 20 minutes earlier	Rejected — `SESSION_EXPIRED`
Delete an attempt without signing in	Refused — "Sign in to manage your attempts"
Call the purge function from the web	Permission denied
Read the demographics table	Present (Sprint 3 migration applied)
Read sessions without signing in	No rows returned (row-level security)

In addition, a real account completed two attempts on the live site and received the keep/delete choice (Figure 2).



Figure 9. LIVE Sprint 4 migration completing in the Supabase SQL editor on the production database.

Acceptance criteria

Each acceptance criterion derived from the client's Sprint 4 wording, with its evidence.

#	CRITERION	HOW IT IS MET	EVIDENCE	RESULT
1	**Create a UI/dashboard for the user**	Dashboard page for signed-in participants	§05, Fig. 1	MET
2	**Store and display data for up to 3 years**	Three-year window; automatic purge	§05, DB tests	MET
3	**Compare with previous years / past attempts, category-wise**	Comparison selector; per-category rows	§07, Fig. 4	MET
4	**Handle previous and current attempts properly**	Newest active; previous kept; two per year	§06, Fig. 2, live	MET
5	**Option to delete/override the previous attempt**	At submission and on the dashboard; replace flag	§06, Fig. 2–3, DB tests	MET
6	**If deleted, only the current attempt is active**	Active result undeletable; others removed	§06, Fig. 3b, DB tests	MET
7	**Compare year-to-year or past data**	Year-to-year chart and table	§07, Fig. 5	MET
8	**Show improved, decreased or stayed similar**	Trend labels with a ±2-point band	§07, Fig. 4	MET
9	**Display using graphs/charts on the dashboard**	Comparison bar chart; year-to-year line chart	§07, Fig. 4–5	MET
10	**Each evaluation has a 10-minute session**	Visible countdown; server time check	§08, Fig. 6, live	MET
11	**Manage what happens when a session expires**	Warnings; answers cleared; nothing stored; restart	§08, Fig. 6	MET
12	**Anonymous while the session is in progress**	No sign-in or writes during the evaluation	§09, network trace	MET
13	**Store data only after the session is completed**	Single transactional submission	§09–10, DB tests	MET

14

Risks and limitations

Known risks carried into handover, with the mitigation in place or planned.

RISK	CONSEQUENCE	MITIGATION	STATUS
Session start time is reported by the browser	A technically skilled participant could extend a session.	The server rejects late submissions. Exam-grade enforcement would need a server-issued token, which conflicts with R6 for anonymous participants.	ACCEPTED
Free-tier database pauses after seven idle days	Sign-in and submission fail while paused.	Daily keep-alive request; answers stay on the device until submission; sign-in shows a clear outage message.	OUTSTANDING
No automatic backups on the free plan	Data loss on project failure is unrecoverable.	Regular CSV export from the admin dashboard, or a paid plan.	OUTSTANDING
Anonymous attempts have no owner	An anonymous participant's data cannot be located for deletion on request.	A consequence of R6. Anonymous rows expire under the three-year purge.	ACCEPTED
Credentials recorded in the handover notes	Anyone holding the notes has full access.	Rotate keys, tokens and admin passwords before handover; the notes are excluded from the repository.	REQUIRED
Email sender not yet configured	Verification codes and results emails may not arrive.	The function is provider-independent; configure the project sender (Section 15).	OUTSTANDING
Front-end libraries load from public CDNs	A CDN outage affects charts and PDF export.	Versions are pinned; self-hosting is a candidate for handover.	ACCEPTED
Build and deployment knowledge held by one team member	Slow recovery if that person is unavailable.	Written handover guide planned for Sprint 5.	SPRINT 5

15

Deployment and outstanding items

Releases

COMMIT	DATE	CHANGE
`98f6586`	13 Sep 2026	Home page redesign; initial attempt rules and session handling; yearly results page
`f9abeef`	13 Sep 2026	User dashboard, current/previous attempts, ten-minute sessions, store on completion
`9ec0cf3`	13 Sep 2026	Redesign of every page on the shared design system
`7646f38`	13 Sep 2026	Refined depth and branding
`8a11b51`	14 Sep 2026	Results email function hardened
`ca23387`	14 Sep 2026	Email transport made provider-independent (SMTP or API)

Front-end releases deploy through GitHub Pages from `main`; asset URLs carry a version parameter so browsers load the new files. Database changes were applied in the Supabase SQL editor. One deployment was delayed by a GitHub Pages service incident and was re-triggered once the service recovered.

Outstanding items

ITEM	DETAIL	STATUS
Email sender configuration	Configure a dedicated project email account as the Supabase authentication SMTP sender and as the edge-function secrets, then deploy the updated `send-results-email` function. Until then verification codes and results emails are not reliably delivered to all participants.	OUTSTANDING
Sign-up trigger fix	Confirm `fix-new-user-trigger.sql` has been applied to the live database.	VERIFY
Credential rotation	Rotate the keys, tokens and admin passwords recorded in the handover notes before handover.	REQUIRED
Database pausing	Add a daily keep-alive request so the free project does not pause again.	OUTSTANDING
Backups	The free plan has no automatic backups; export participant data (CSV) regularly or move to a paid plan.	OUTSTANDING
Ownership	Transfer the GitHub repository and Supabase project to university-owned accounts.	SPRINT 5
PDF export check	Re-test the report page's PDF export on the redesigned page.	VERIFY
Team details	Following a change in team membership, the home page team section now matches this report (Product Owner Albin Roy; Tester Merajun Nabi). Ships with the next front-end release.	NEXT RELEASE

> **CONSIDERATIONS FOR THE NEXT SPRINT**
>
> Email from a university sender address; handover of accounts with rotated credentials; a written handover guide (setup, migration order, administrators, recovery from pausing); and research views that break scores down by demographic group.

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Retrospective

What the team keeps, what it changes, and the actions carried into Sprint 5.

What worked

- Putting the evaluation rules in the database behind a single write path. Every rule held under the live probes, and the browser could be simplified because it no longer had to be trusted.

- Testing the migration in PGlite before touching production. The suite caught the draft that would have deleted same-year attempts (Section 12, defect 1).
- Verifying R6 with a network trace rather than by reading the code. It showed the anonymous sign-in and early session row that the code review had missed.
- A shared design system. Seven pages were rebuilt in one release with a consistent look, and the admin dashboard inherited it for free.

What to change

- Two requirements (R3, R5) were first built to an interpretation and then rebuilt to the client's wording. Confirm the wording in writing at planning before any migration is drafted.
- The production database paused without anyone noticing. A live service needs a monitor, not a periodic manual check.
- The team roster on the home page drifted from the sprint documents. Keep one source for team details and copy from it.
- Sprint reports and slides were briefly published with the site. Documents that are not part of the product stay out of the published tree.

Actions for Sprint 5

ACTION	WHY
Configure the project email sender and redeploy the results function	Closes the only outstanding requirement-level item (S11).
Add a keep-alive request and a scheduled data export	Removes the two failure modes of the free tier (Section 14).
Rotate every credential and hand ownership to university accounts	Required before the platform leaves the team.
Write the handover guide	Setup, migration order, administrators and recovery from pausing, so the platform does not depend on one person.

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Appendix — files added and modified

FILE	CHANGE	PURPOSE
`sprint4-attempts.sql`	Added	Attempt rules, session limit, store on completion, retention
`my-results.html`, `js/my-results.js`, `css/my-results.css`	Added	User dashboard
`js/session.js`	Added	Identity and ten-minute session clock
`js/landing.js`	Added	Answerable first statement on the home page
`supabase/templates/confirm-signup.html`	Added	Verification-code email template
`js/assessment.js`, `js/questionnaire.js`	Modified	Submission, history, timer, expiry, attempt choice
`js/auth.js`	Modified	Anonymous start without sign-in; clearer errors
`js/admin.js`, `admin.html`, `css/admin.css`	Modified	Redesign; active attempts only in averages
`js/results.js`, `js/report.js`, `js/questions.js`	Modified	Chart colours and redesign
`css/main.css` and page stylesheets	Modified	Shared design system
All page HTML files	Modified	Redesign, session script, versioned assets
`supabase/functions/send-results-email/index.ts`	Modified	Hardened, provider-independent results email
`js/cursor.js`	Removed	Decorative custom cursor
`fix-new-user-trigger.sql`, `demo-sprint4-seed.sql`	Added (local)	Account-creation fix; demonstration history for test accounts